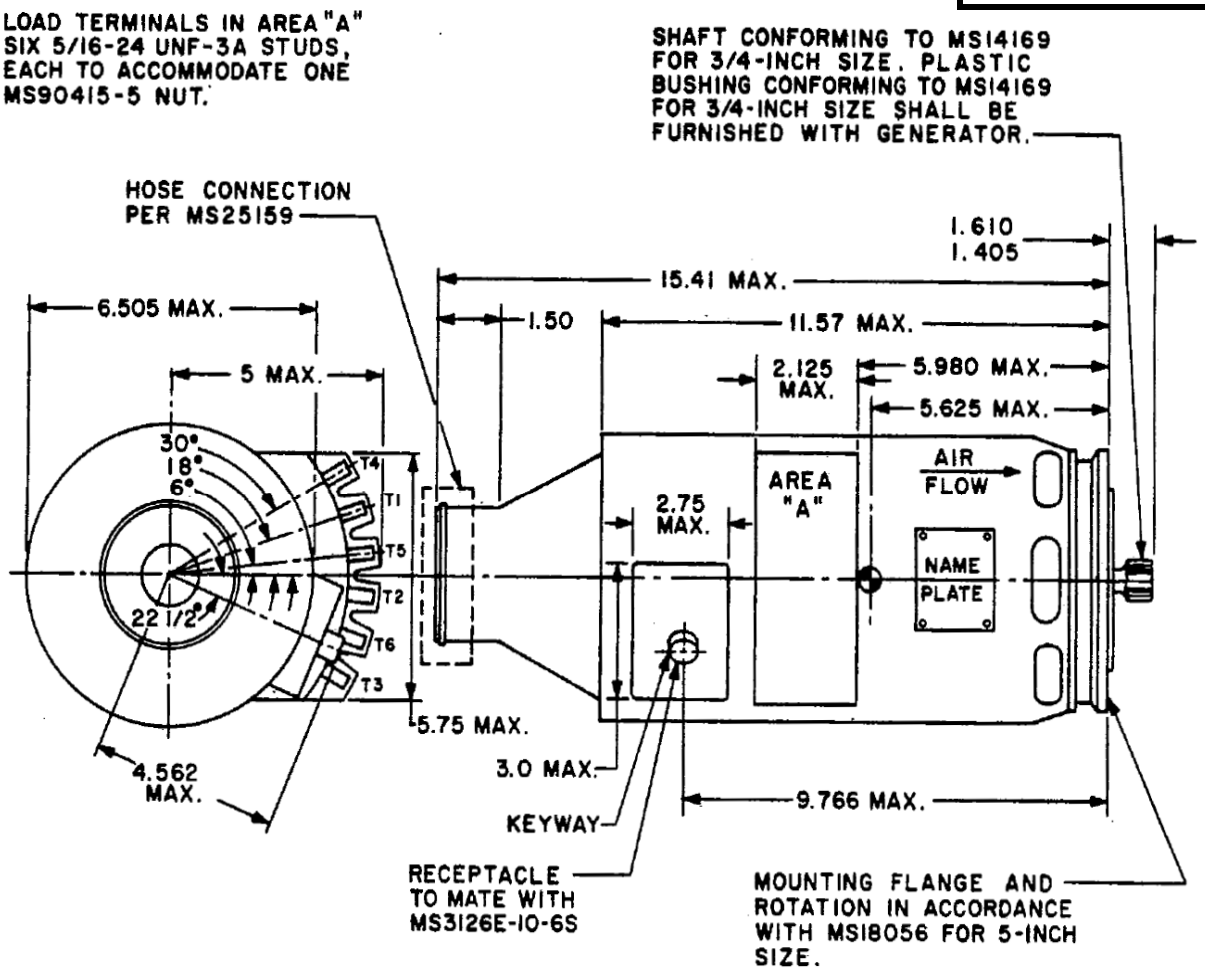
FED. SUP CLASS
6115



GENERATOR DATA	
RATED VOLTAGE	120/208
RATED OUTPUT, KVA	20
PHASES	3
SPEED RANGE, RPM	7600-8400
MAX. SPEED FOR REGULATION, RPM	10000
OVERSPEED, RPM	11000
RATED POWER FACTOR	0.75 TO 1.0
EFFICIENCY, MIN. AT RATED LOAD, PCT	86 %
MAX. GENERATOR WEIGHT, POUNDS	53
OVERHUNG MOMENT, MAX, IN-POUNDS	290
FREQUENCY, HERTZ	380-420
SHEAR, INCH-POUNDS	1000-1400
FLEXIBLE DRIVE	REQUIRED
GEN. ROTOR MOMENT OF INERTIA	WR^2-0.3 LB-FT^2

GENERATOR
P/N MS90304-3
SUPERSEDES MS90304-1

(E) ENTIRE STANDARD REVISED

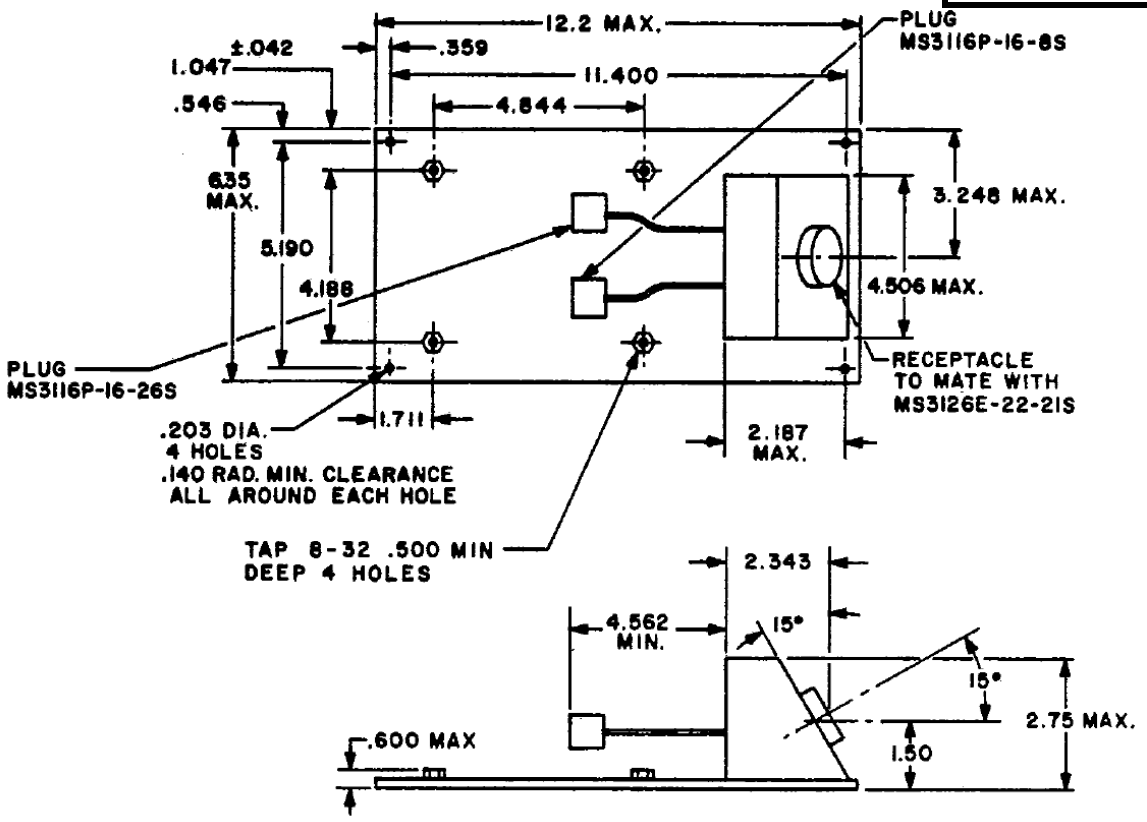
APPROVED 28 APRIL 1964 REVISED(A) 5 OCT. 1965 (B) 27 SEPT 1970 (C) 12 DEC 1975 (D) 10 NOV 1978 (E) 24 SEPT 79

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P.A. NAVY – AS Other Cust.	TITLE GENERATOR SYSTEM, 20 KVA, 400 HERTZ, ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS, AIRCRAFT, SELF-COOLED	MILITARY STANDARD MS90304(AS)
PROCUREMENT SPECIFICATION MIL-G-21480	SUPERSEDES	SHEET 1 OF 9

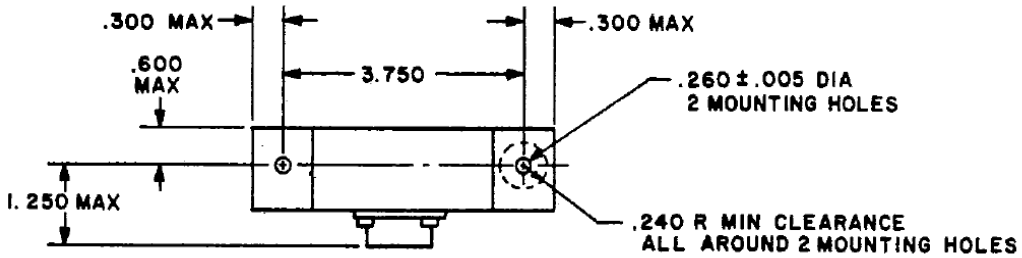
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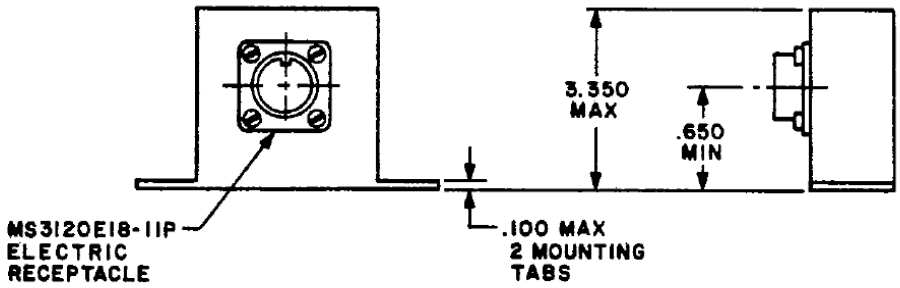
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BASE

PART NUMBER MS90304-9. MAXIMUM WEIGHT 2.5 POUNDS





JUMPER BOX

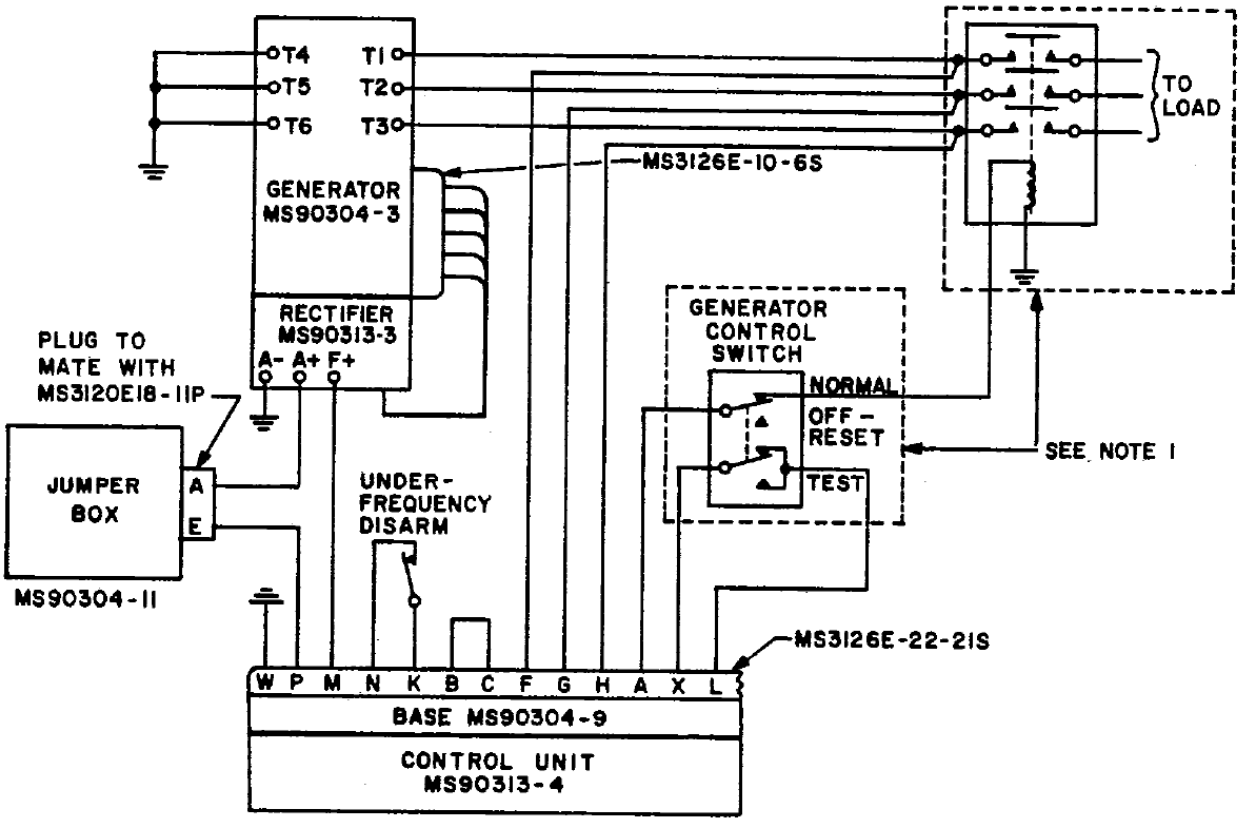
P/N MS90304-11

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PROCUREMENT SPECIFICATION MIL-G-21480	SUPERSEDES	SHEET 2 OF 9

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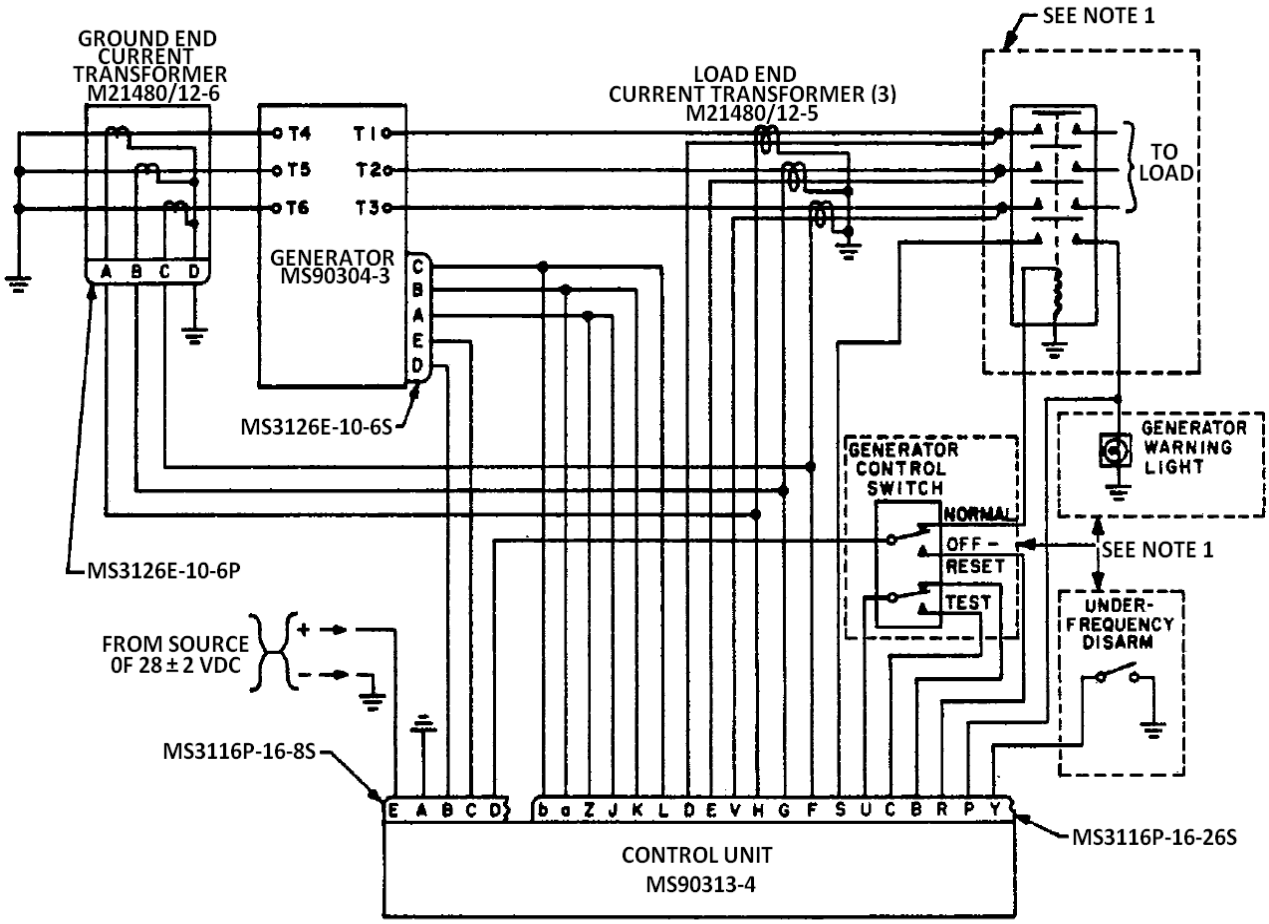
TEST CIRCUIT FOR MS90304-10 SYSTEM

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PROCUREMENT SPECIFICATION MIL-G-21480	SUPERSEDES	SHEET 3 OF 9

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TEST CIRCUIT FOR MS90304-20 SYSTEM

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PROCUREMENT SPECIFICATION MIL-G-21480	SUPERSEDES	SHEET 4 OF 9

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REQUIREMENTS:

1. THE AIRCRAFT MANUFACTURER SHALL ALLOW CONNECTING LEAD LENGTH AND BLAST TUBE LENGTH FOR THE LIMITING GENERATOR DIMENSIONS SHOWN ON THIS DRAWING.

2. THE AIRCRAFT MANUFACTURER SHALL PROVIDE CONNECTING LEADS SHOWN.

3. THE AIRCRAFT MANUFACTURER SHALL FURNISH MAXIMUM CONTINUOUS CURRENT AND LIMITING RESISTANCE VALUE FOR EACH INTERCONNECTING WIRE.

4. THE AIRCRAFT MANUFACTURER SHALL ALLOW ADEQUATE CLEARANCE FOR INSTALLING AND REMOVING THE ITEMS SHOWN ON THIS STANDARD FROM THE AIRCRAFT.

5. THE REQUIREMENTS OF MIL-G-21480 PERTAIN WITH THE FOLLOWING EXCEPTIONS:

DELETE THE FOLLOWING PARAGRAPHS:

3.4.2.2 THROUGH 3.4.2.8, 3.4.10, 3.4.10.5, 3.4.10.6, 3.4.13.1, 3.4.14 THROUGH 3.4.16, 3.4.16.1 THROUGH 3.4.16.5, 3.5.7.5, 4.4.1.2.1, 4.4.2.3, 4.5.20, 4.5.22.1 THROUGH 4.5.22.3, 4.5.22.3.1 THROUGH 4.5.22.3.4, 4.5.24.1 THROUGH 4.5.24.3, 4.5.24.3.1 THROUGH 4.5.24.3.4, 4.5.25.1, 4.5.25.2, 4.5.25.2.1, 4.5.28, 4.5.32, 4.5.33, 4.5.33.1, 4.5.33.2.

3.3.3.4 ADD: GENERATOR. THE GENERATOR HOUSING SHALL BE MADE OF STEEL CONFORMING TO THE REQUIREMENTS OF FED-STD-66. THE SURFACE FINISH SHALL BE IN ACCORDANCE WITH QQ-P-416, CLASS 2, TYPE II.

3.4.2 DELETE AND ADD: ENVIRONMENTAL REQUIREMENTS. THE SYSTEM SHALL MEET THE REQUIREMENTS OF MIL-E-81910 UNLESS OTHERWISE SPECIFIED BY THE SPECIFICATION SHEET.

3.4.2.1 DELETE AND ADD: TEMPERATURE AND ALTITUDE. THE GENERATOR SHALL MEET THE TEMPERATURE-ALTITUDE REQUIREMENTS OF FIGURE 1 OF THE SPECIFICATION SHEET WHILE OPERATING UNDER THE SELF-COOLED CONDITIONS. THE CONTROL COMPONENTS SHALL OPERATE CONTINUOUSLY WHEN SUBJECTED TO THE TEMPERATURE-ALTITUDE REQUIREMENTS OF MIL-E-81910 FOR SELF-COOLED EQUIPMENT.

ALTITUDE FEET	INLET AIR TEMP. DEGREES C	GENERATOR MAXIMUM CONTINUOUS LOAD - KVA	
		.75 POWER FACTOR	UNITY POWER FACTOR
O-SEA LEVEL	40	28	30
O-SEA LEVEL	20	30	32
O-SEA LEVEL	10	32	32
O-SEA LEVEL	0	32	34
O-SEA LEVEL	-20	34	34
15,000	20	24	24
15,000	10	24	26
15,000	0	24	26
15,000	-20	26	28
25,000	20	20	20
25,000	10	20	20
25,000	0	20	22
25,000	-20	22	24

FIGURE 1. GENERATOR TEMPERATURE-ALTITUDE REQUIREMENTS (SELF-COOLED)

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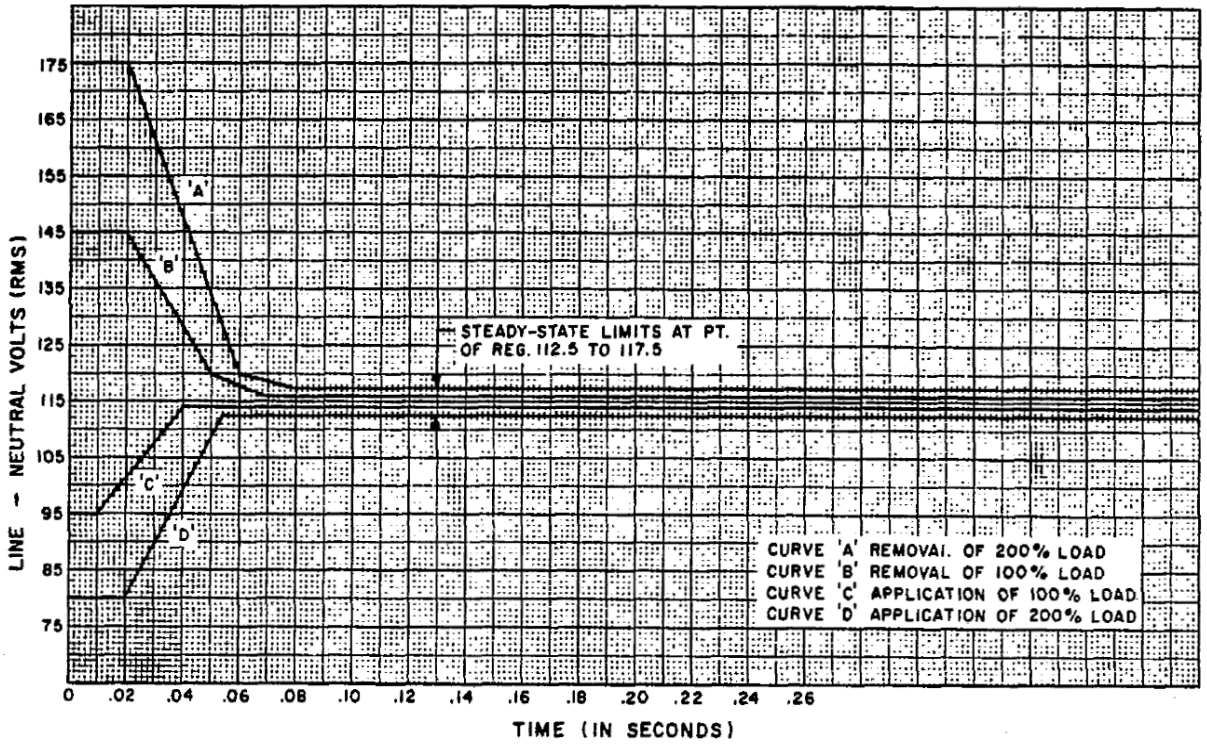


FIGURE 2. TRANSIENT AND STEADY-STATE VOLTAGE LIMITS UPON APPLICATION AND REMOVAL OF LOADS.

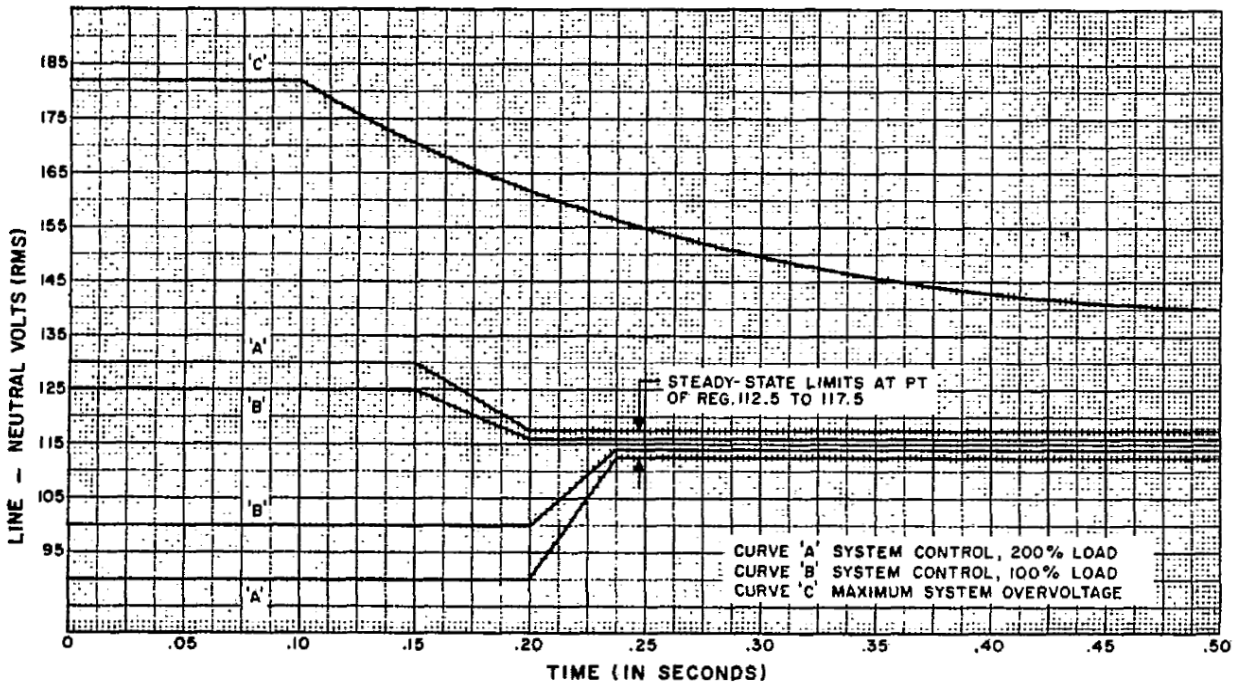


FIGURE 3. TRANSIENT AND STEADY-STATE VOLTAGE LIMITS UPON CLOSURE OF THE GENERATOR CONTROL SWITCH

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PROCUREMENT SPECIFICATION MIL-G-21480	SUPERSEDES	SHEET 6 OF 9

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3.4.8 CHANGE "FIGURE 3" TO "FIGURE 2 OF THE SPECIFICATION SHEET".

3.4.9.3 ADD: THE GENERATOR SYSTEM SHALL BE DESIGNED SO THAT UPON REMOVAL OF ANY ELECTRIC PLUG THE GENERATOR AND GCU SHALL BE DE-ENERGIZED.

3.4.9.6 ADD: THE GENERATOR SHALL BE SELF-COOLED AND IN ADDITION SHALL BE PROVIDED WITH MEANS FOR INSTALLATION OF AN AIR INLET TUBE.

3.4.9.6.1.1 CHANGE "10,000 HOUR LIFE" TO "2500 HOUR LIFE".

3.4.9.7 CHANGE "10,000 HOURS" TO "AT LEAST 2500 HOURS:

3.4.9.8.1 ADD: RECTIFIER ASSEMBLY. IN THE MS90304-10 SYSTEM, THE GENERATOR SHALL BE EQUIPPED WITH AN EXTERNALLY MOUNTED RECTIFIER CONFORMING TO MS90313-3 FOR RECTIFICATION OF PMG OUTPUT TO THE GENERATOR EXCITER FIELD.

3.4.10.3 CHANGE "5000 HOURS" TO "4000 HOURS".

3.4.17 DELETE AND ADD: RELIABILITY. THF SYSTEM MTBF (MEAN TIME BETWEEN FAILURE) SHALL BE NOT LESS THAN 1200 HOURS. THIS FIGURE SHALL APPLY TO FAILURES WHICH WILL CAUSE SYSTEM OPERATION TO DEVIATE FROM PERFORMANCE REQUIREMENT.

3.5.1 CHANGE "FIGURE 3" TO "FIGURES 2 AND 3 OF THE SPECIFICATION SHEET".

3.5.4 DELETE AND ADD: WAVEFORM. THE CREST FACTOR FOP EACH PHASE VOLTAGE WAVE FORM SHALL BE 1.41 ± 5%. NO SINGLE HARMONIC SHALL EXCEED 3% OF THE FUNDAMENTAL.

3.5.5 CHANGE "5 AMPS MINIMUM CONTINUOUS CAPACITY" TO "3 AMPS MINIMUM CONTINUOUS CAPACITY".

3.5.7.2 CHANGE "FIGURE 6" TO "FIGURE 3 OF THE SPECIFICATION SHEET".

3.5.7.3 DELETE AND ADD: UNDERVOLTAGE. UNDERVOLTAGE PROTECTION SHALL BE PROVIDED BY A STATIC ELEMENT WHICH FUNCTIONS TO DISCONNECT THE SYSTEM FROM THE LOAD BUS AND DEENERGIZE THE GENERATOR WITHIN 6 + 1 SECONDS AFTER ANY PHASE VOLTAGE (LINE-NEUTRAL) FALLS TO 100 + 5 VOLTS. THE SYSTEM SHALL BE CAPABLE OF BEING RESET WITHIN 2 SECONDS AFTER THE LOWEST PHASE VOLTAGE EXCEEDS 105 VOLTS.

3.5.7.4 DELETE AND ADD: UNDERFREQUENCY. UNDERFREQUENCY PROTECTION SHALL BE PROVIDED BY A STATIC ELEMENT WHICH FUNCTIONS TO DISCONNECT THE SYSTEM FROM THE LOAD BUS WITHIN 2 TO 5 SECONDS AFTER THE FREQUENCY FALLS TO 375 + 5 HZ. AFTER EITHER AUTOMATIC OR MANUAL RESET, THE UNDERFREQUENCY PROTECTION SHALL PROVIDE FOR THE SYSTEM TO BE AUTOMATICALLY RECONNECTED TO THE BUS WITHIN 2 SECONDS AFTER THE FREQUENCY EXCEEDS 378 ± 3 HZ. THE SPREAD BETWEEN DROPOUT AND PICKUP SHALL NOT BE LESS THAN 4 HZ.

3.5.7.6 DELETE AND ADD: FEEDER FAULT. THE MS90304-20 SYSTEM SHALL PROVIDE FOR THE GENERATOR TO BE DEENERGIZED AND DISCONNECTED FROM THE LOAD BUS WHEN THE FAULT CURRENT ON ANY PHASE IS 18 ± 3 AMPERES.

3.5.8 DELETE AND ADD: ELECTROMAGNETIC INTERFERENCE. THE SYSTEM SHALL MEET THE REQUIREMENTS OF MIL-E-81910, CLASS NO. IIIB FOR THE GENERATORS AND CONTROL EQUIPMENT, EXCEPT THE TEN MICROFARAD FEEDTHROUGH CAPACITOR SHALL BE REMOVED.

4.4.2.1 ADD: QUALIFICATION TESTS SHALL BE SUPPLEMENTED WITH A MINIMUM 100 HOUR FAILURE FREE FLIGHT TEST BEFORE CONSIDERATION IS GIVEN TO INCORPORATING THE ITEM ON THE QUALIFIED PRODUCTS LIST.

4.4.2.2 ADD: THE GENERATOR SHALL BE SELF-COOLING, NO EXTERNAL BLAST AIR PROVIDED.

4.4.2.3 ADD: THE GENERATOR SHALL BE SELF-COOLING, NO EXTERNAL BLAST AIR PROVIDED.

4.5.3.1 CHANGE "FIGURE 3" TO "FIGURE 3 OF THE SPECIFICATION SHEET".

4.5.3.2 CHANGE "FIGURE 3" TO "FIGURE 3 OF THE SPECIFICATION SHEET".

4.5.9 DELETE PART (D) ADD:

(D) AT MINIMUM, MAXIMUM AND AVERAGE RATED GENERATOR RPM. THE EFFICIENCY AT RATED LOAD SHALL NOT BE LESS THAN 86%. THE EFFICIENCY SHALL NOT BF LESS THAN 84% UNDER ANY CONDITION GREATER THAN 1/2 RATED CURRENT. THE EFFICIENCY AT 1/4 RATED CURRENT SHALL NOT BE LESS THAN 78%.

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4.5.12 DELETE AND ADD: ELECTROMAGNETIC INTERFERENCE. THE SYSTEM SHALL BE SUBJECTED TO AN ELECTROMAGNETIC INTERFERENCE TEST IN ACCORDANCE WITH MIL-E-81910, EXCEPT THE 10 MFD FEED-THROUGH CAPACITOR SHALL BE REMOVED.

4.5.13 DELETE PART (C) AND ADD:

(C) CONDITIONS (A) AND (B) SHALL BE REPEATED AT AVERAGE AND MAXIMUM RATED SPEED. THE LINE-NEUTRAL CREST FACTOR OBTAINED BY THE ABOVE TEST SHALL BE 1.41 ± 5%. THE VALUE OF ANY HARMONIC DURING THE ABOVE TEST SHALL NOT EXCEED 3% OF THE FUNDAMENTAL.

4.5.20.1 DELETE AND ADD: SALT FOG, CONTROL COMPONENTS. CONTROL COMPONENTS SHALL BE SUBJECTED TO A SALT FOG TEST IN ACCORDANCE WITH MIL-E-81910, EXCEPT THE DURATION OF THE TEST SHALL NOT EXCEED 25 HOURS. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD, AND PROTECTIVE FUNCTIONS.

4.5.20.2 DELETE AND ADD: OIL-SALT WATER INGESTION, GENERATORS. THE GENERATOR SHALL BE SUBJECTED TO AN OIL-SALT WATER INGESTION TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS AND THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD.

4.5.21 DELETE AND ADD: FUNGUS RESISTANCE. SYSTEM COMPONENTS SHALL BE SUBJECTED TO A FUNGUS TEST IN ACCORDANCE WITH MIL-E-81910, WITH THE INSTALLATION INSTRUCTIONS AND MOUNTING NUTS (AS REQUIRED BY PARAGRAPH 4.3.1) ATTACHED. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD, AND PROTECTIVE FUNCTIONS. MOUNTING NUTS ARE TO BE UNAFFECTED AND THE PACKAGE HOLDING THE NUTS SUITABLE FOR SHIPMENT. INSTRUCTIONS ARE TO BE LEGIBLE.

4.5.22 DELETE AND ADD: SAND AND DUST. THE GENERATOR SHALL BE SUBJECTED TO A SAND AND DUST TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS AND THE EFFECTS OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD.

4.5.23.1.1 DELETE AND ADD; GENERATOR. THE GENERATOR AND RECTIFIER ASSEMBLY SHALL BE SUBJECTED TO A VIBRATION TEST IN ACCORDANCE WITH MIL-E-81910 FOR ENGINE-MOUNTED COMPONENTS. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS AND THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD.

4.5.23.1.2 DELETE AND ADD: CONTROL COMPONENTS. CONTROL COMPONENTS SHALL BE SUBJECTED TO A VIBRATION TEST IN ACCORDANCE WITH MIL-E-81910, EXCEPT THAT THE VIBRATION RANGE SHALL BE FROM 5 TO 500 HZ. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD, AND PROTECTIVE FUNCTIONS.

4.5.24 DELETE AND ADD: SHOCK. SYSTEM COMPONENTS SHALL BE SUBJECTED TO A SHOCK TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF PROCEDURE I, THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD, AND PROTECTIVE FUNCTIONS. THERE SHALL BE NO FAILURE OF THE MOUNTING ATTACHMENTS, AND THE TEST ITEM SHALL REMAIN IN PLACE AND NOT CREATE A HAZARD DURING PROCEDURE III.

4.5.25 DELETE AND ADD: HUMIDITY. SYSTEM COMPONENTS, EXCEPT THOSE TESTED UNDER PARAGRAPH 4.5.20.2, SHALL BE SUBJECTED TO A HUMIDITY TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD, AND PROTECTIVE FUNCTIONS.

4.5.30 DELETE FIRST SENTENCE AND ADD: ALL SYSTEM COMPONENTS EXCEPT THE GENERATOR SHALL BE SUBJECTED TO THE ACCELERATION TESTS OF MIL-E-81910. AT THE COMPLETION OF THE TESTS THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILDUP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF RATED AND 200% LOAD, AND PROTECTIVE FUNCTIONS.

NOTES:

1. GENERATOR CONTROL SWITCHES, WARNING LIGHTS, AND LINE CONTACTORS ARE SHOWN FOR INTERCONNECTIONS ONLY AND ARE NOT FURNISHED UNDER THIS DRAWING.

2. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: LINEAR 0.030, ANGULAR ± 1°.

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IDENTIFICATION OF ITEMS	
ITEM	MILITARY PART NUMBER
FIELD-INSTALLED SYSTEM (CONSISTS OF ONE GENERATOR, ONE RECTIFIER, ONE CONTROL UNIT, ONE BASE, AND ONE JUMPER BOX)	MS90304-10
FACTORY-INSTALLED SYSTEM (CONSISTS OF ONE GENERATOR, ONE CONTROL UNIT, THREE LOAD END CURRENT TRANSFORMERS, AND ONE GROUND END CURRENT TRANSFORMER)	MS90304-20
GENERATOR	MS90304-3
RECTIFIER	MS90313-3
CONTROL UNIT	MS90313-4
BASE	MS90304-9
JUMPER BOX	MS90304-11
LOAD END CURRENT TRANSFORMER	M21480/12-5
GROUND END CURRENT TRANSFORMER	M21480/12-6

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